EMBL's European Bioinformatics Institute (EMBL-EBI)

Automating Galaxy: Enhancing Production Pipelines with Galaxy-wf-executor

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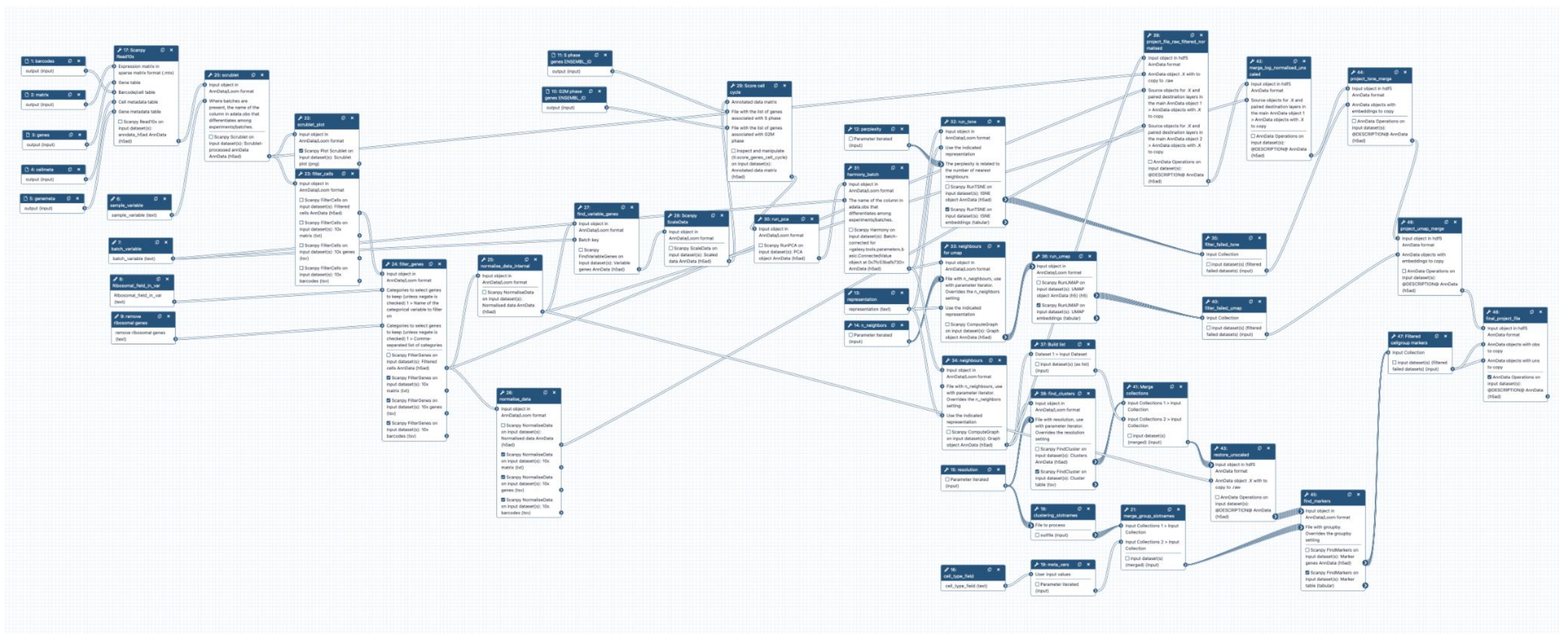
Introduction

Galaxy is known as a user-friendly workflow system for biologists without significant computational expertise. It has evolved to also support large-scale bioinformatics production pipelines for major data providers and core facilities.

Workflows in Galaxy are generally executed using Galaxy web user-interface and in that scheme are not an optimal solution when it comes to production usage.

Here, we present Galaxy-wf-executor, a fully automated way of executing workflows on Galaxy. It is based on Bioblend and it manages data transfers, configures histories and workflows, schedules and monitors workflows, and retrieves results.

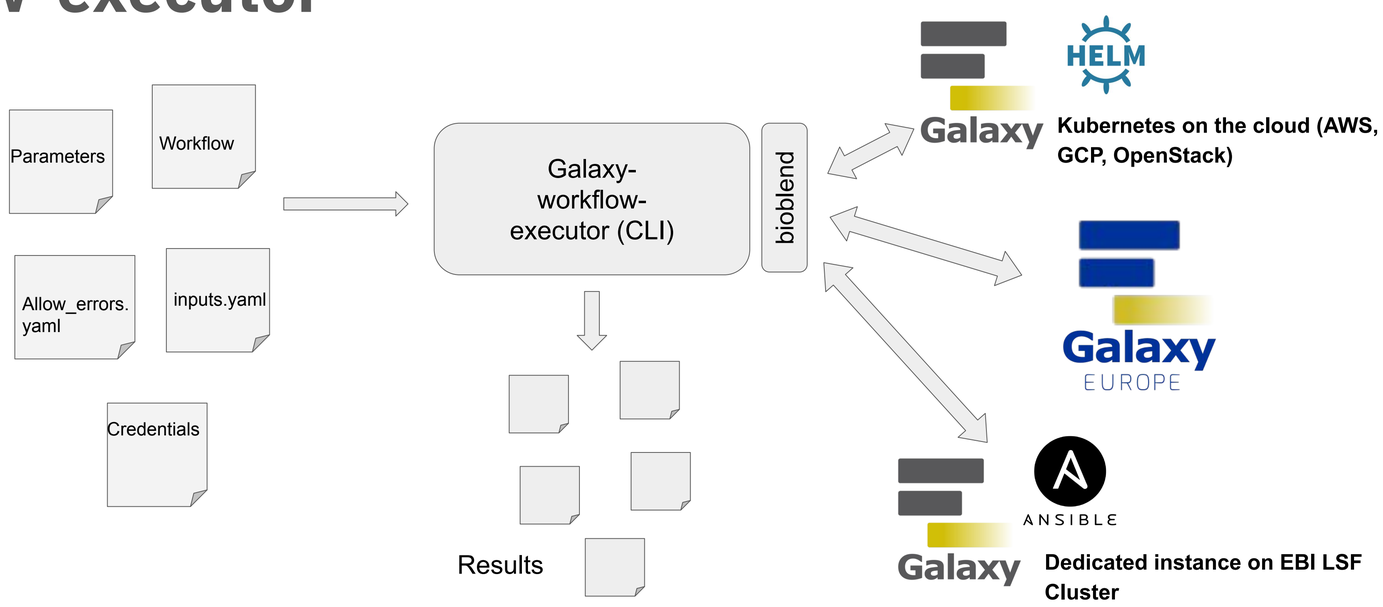
Example workflow



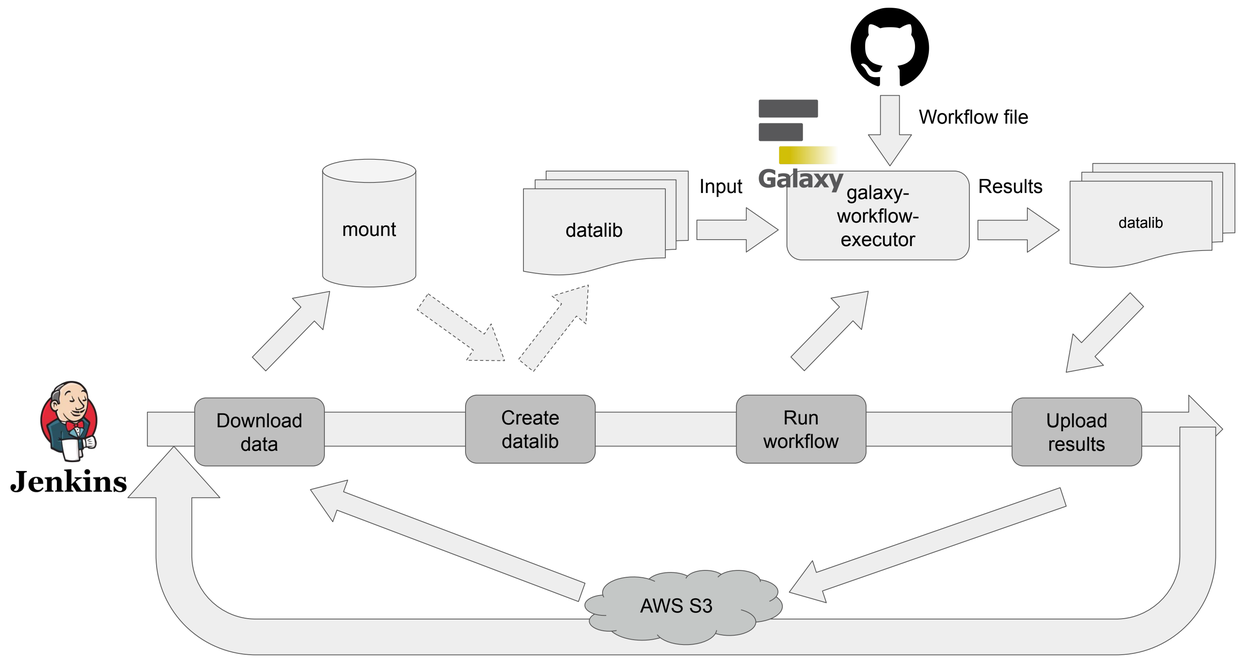
Running Galaxy-workflow-executor

The process begins with uploading datasets and the corresponding workflow. Once uploaded, the workflow is executed, during which the system runs the analysis and waits for the results. Specific steps in the workflow can be configured to allow for errors without halting the entire process. Finally, after the run is complete, the results are retrieved for further analysis.

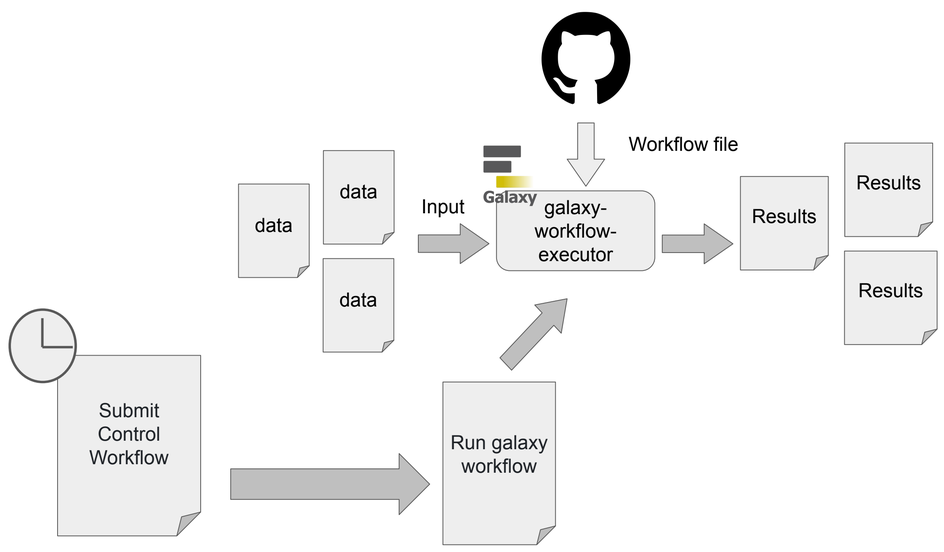
The process generates several results. First, the results are downloaded and can be uploaded to the data library for easy access and future reference. Additionally, the results can be persisted in the history, ensuring they are stored securely and can be reviewed or analyzed at any time.



Use case at Persist-seq



Use case at EMBL-EBI



Bottleneck

The runners needed modifications to be compatible with different schedulers (LSF, SLURM, SGE, Kubernetes) and the resubmission setup. An out-of-the-box configuration that can implement resubmission logic across runners would be advantageous. This could include a templated resubmission setup, potentially through the TPV module.

Open-source

The galaxy-workflow-executor with all underlying dependencies, including scripts, are available through Bioconda and Biocontainers, and Galaxy tools are available via the Galaxy ToolShed.

Galaxy ToolShed

BIOCONDA GitHub

Galaxy workflow executor